

Assignment II Engineering Thermodynamics II

Submission Date: 06/04/2018 E.C

Assignments must be submitted in the same groups as the first assignment.

Solve the following problems, showing all necessary steps clearly and neatly.

1. A 100-m³ storage tank with fuel gases is at 20°C, 100 kPa containing a mixture of acetylene (C₂H₂), propane (C₃H₈), and butane (C₄H₁₀). A test shows that the partial pressure of the C₂H₂ is 15 kPa and that of the C₃H₈ is 65 kPa. How much mass is there of each component?
2. A rigid container has 1 kg CO₂ at 300 K and 1 kg Ar at 400 K, both at 150 kPa. Now they are allowed to mix without any heat transfer. What is the final T, P?
3. Atmospheric air is at 100 kPa, 25°C and relative humidity 65%. Find the absolute humidity and the dew point of the mixture. If the mixture is heated to 35°C, what is the new relative humidity?
4. Consider 100 m³ of atmospheric air which is an air–water vapor mixture at 100 kPa, 15°C and 40% relative humidity. Find the mass of water and the humidity ratio. What is the dew point of the mixture?
5. Consider a 1-m³/s flow of atmospheric air at 100 kPa, 25°C, and 80% relative humidity. Assume this mixture flows into a basement room where it cools to 15°C, 100 kPa. Find the rate of water condensing out and the exit mixture volume flow rate?
6. Air enters a 30-cm-diameter cooling section at 1 atm, 35 °C, and 45 percent relative humidity at 18 m/s. Heat is removed from the air at a rate of 750 kJ/min. Determine (a) the exit temperature, (b) the exit relative humidity of the air, and (c) the exit velocity
7. Air enters a 40-cm-diameter cooling section at 1 atm, 32 °C, and 70 percent relative humidity at 120 m/min. The air is cooled by passing it over a cooling coil through which cold water flows. The water experiences a temperature rise of 6 °C. The air leaves the cooling section saturated at 20 °C. Determine (a) the rate of heat transfer, (b) the mass flow rate of the water, and (c) the exit velocity of the airstream.
8. Air enters an evaporative cooler at 1 atm, 40 °C, and 20 percent relative humidity at a rate of 7 m³/min, and it leaves with a relative humidity of 90 percent. Determine (a) the exit temperature of the air and (b) the required rate of water supply to the evaporative cooler.
9. A stream of warm air with a dry-bulb temperature of 36 °C and a wet-bulb temperature of 30 °C is mixed adiabatically with a stream of saturated cool air at 12 °C. The dry air mass flow rates of the warm and cool airstreams are 8 and 10 kg/s, respectively. Assuming a total pressure of 1 atm, determine (a) the temperature, (b) the specific humidity, and (c) the relative humidity of the mixture.
10. A wet cooling tower is to cool 25 kg/s of cooling water from 40 to 30 °C at a location where the atmospheric pressure is 96 kPa. Atmospheric air enters the tower at 20 °C and 70 percent relative humidity and leaves saturated at 35 °C. Neglecting the power input to the fan, determine (a) the volume flow rate of air into the cooling tower and (b) the mass flow rate of the required makeup water